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Supplementary materials outline

The supplementary materials are organized as follows. We discuss limitations of our study in [Appendix A](#) and potential broader implications of our work in [Appendix B](#). In [Appendix C](#) we provide details of the LLMs used and their hyperparameters, details of our systematic dataset review, dataset filtering and implementation in **AbstentionBench**, and implementation of the LLM judges. In [Appendix D](#) we present additional results for our experiments. In [Appendix E](#) we present a fast subset of the benchmark and discuss code and reproducibility. Finally, in [Appendix F](#) we provide qualitative examples of prompts and model responses.

A Limitations

Given the open-ended nature of dialogue with generative language models, the space of abstention scenarios is broad ranging. While we made a considerable effort to mine and filter hundreds of existing datasets—alongside creating our own underspecified reasoning datasets—there may of course be scenarios where abstention is warranted that we have not covered. We also only focus on English datasets, although future work should also explore abstention in other languages.

Given today’s training datasets can encompass any text available on the web, another limitation is the potential for leakage between evaluation benchmarks and data used in pre- or post-training. For example, the train split of CoCoNot forms part of the Tülu post-training dataset [77], which introduces a confounding factor in abstention performance, since the evaluation now contains examples resembling post-training for the OLMo and Tülu model families. We believe closed or dynamically generated datasets can help address some of these challenges.

Our findings are necessarily restricted to fixed selection of models. We consider 20 leading LLMs with various training paradigm and scales, but of course could only cover a finite and reasonable number. By spanning a range of different models, we hope that our results are sufficiently representative of contemporary model capabilities.

Finally, to evaluate both abstention and correctness, given the numerous and open-ended ways uncertainty can be expressed, we rely on an LLM judge. While we emulate the best practices from prior work, tuning the underlying model and prompt, any LLM judge will be imperfect. To confirm the quality of our overall results we compare the abstention judge responses against human annotations of model responses, and find our judge achieves high performance relative to human ground truth (see [Appendix C](#)).

B Broader impacts

This work is an empirical evaluation of the abstention capabilities of frontier LLMs. We highlight both strengths and weaknesses in state-of-the-art models, including reasoning fine-tuned models, and suggest that failure to abstain may result from a fundamental inability to reason about uncertainty.

We do not foresee the direct application of our benchmark as leading to harm. However, as we note in [Appendix A](#), our work relies on LLM judges, which will introduce some noise into our results. It is possible that such noise could lead to overconfidence in abstention capabilities with respect to certain scenarios, possibly leading to inappropriate deployments.

We hope the release of **AbstentionBench** will encourage further research evaluating and mitigating shortcomings with respect to an important capability. However, as **AbstentionBench** only makes use of publicly available data, and does not retain a private or gated test set, it is possible that performance estimates may become inflated over time due to dataset reuse and overfitting to benchmark idiosyncrasies [84, 85, 86]. Future work may consider a private or gated test sets.